

CYBERSIFT

Technical Assessment Report

Prepared by

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Role Applied For

24/7 Security Analyst

Submitted To

CyberSift

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1. Objective Summary

This report documents the completion of the CyberSift technical assessment for the 24/7 Security Analyst role. The assessment required deploying a centralized security log management environment using OpenSearch, OpenSearch Dashboards, and Apache NiFi on an Ubuntu 22.04 LTS virtual machine, building a functioning syslog ingestion pipeline, and performing threat analysis on provided Windows Event Log (EVTX) files.

Assessment Scope

- Environment Setup — Deploy Ubuntu 22.04 LTS VM, install and configure OpenSearch, OpenSearch Dashboards, and Apache NiFi
- Pipeline Creation — Build a NiFi data flow to receive UDP syslog on port 5514 and route data into OpenSearch for indexing
- Visualization — Create index patterns and dashboards in OpenSearch Dashboards to visualize ingested data
- Threat Analysis — Ingest three provided .evtx Windows Event Log files, identify and document three distinct attack patterns

Environment Summary

Operating System	Ubuntu 22.04.5 LTS (Jammy Jellyfish)
Hostname	Cybershift-VM
Hypervisor	Oracle VirtualBox
Architecture	x86-64
Kernel	Linux 6.8.0-111-generic
OpenSearch	Version 2.19.1
NiFi	Version 1.28.1 (HTTP mode, port 8080)
VM IP Address	10.0.2.15

2. Environment Setup

The assessment environment was built on Ubuntu 22.04.5 LTS running inside an Oracle VirtualBox virtual machine. The following sections document each installation and configuration step performed.

2.1 Virtual Machine Specifications

VM Name	Cybershift-VM
OS	Ubuntu 22.04.5 LTS Desktop
Virtualization	Oracle VirtualBox (innotek GmbH)
Machine ID	e3ef24523d594cd5a30cda67e26a83f9
Kernel	Linux 6.8.0-111-generic x86-64
Network	NAT (IP: 10.0.2.15)

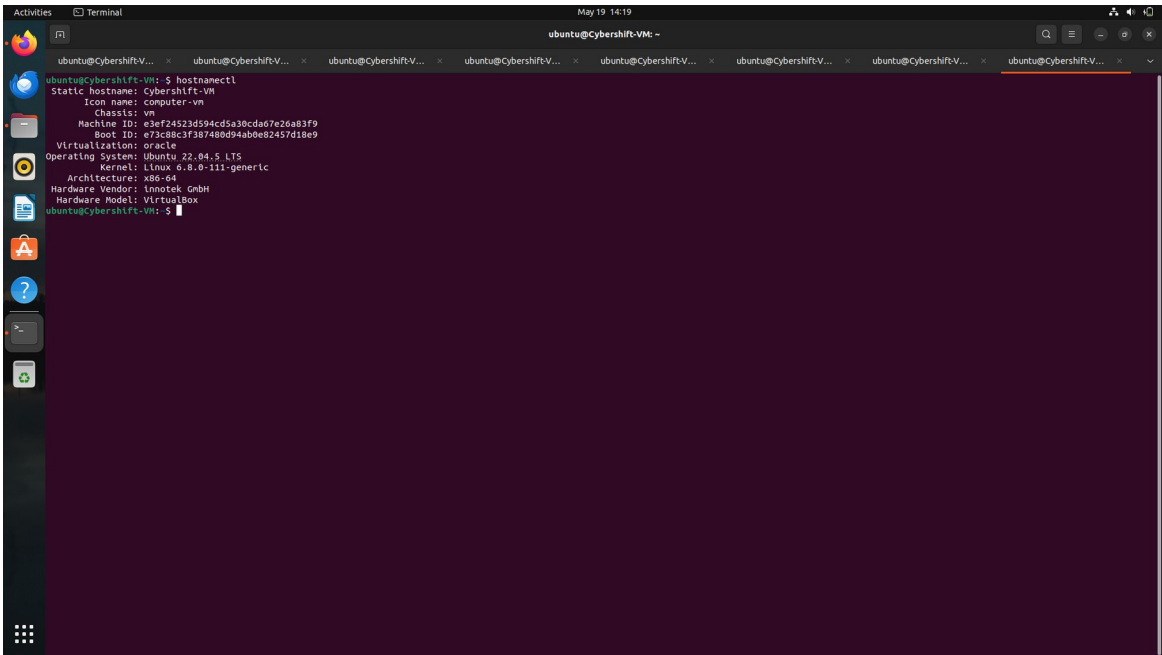
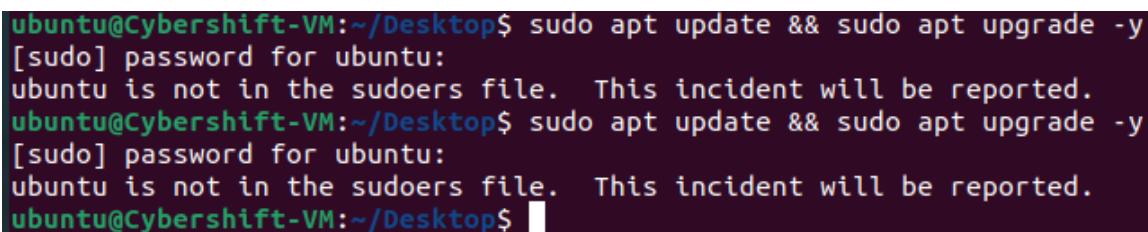


Fig. 1 — hostnamectl: Ubuntu 22.04.5 LTS on VirtualBox

2.2 System Update and Package Prerequisites

Before installing any components, the system package list was updated and upgraded to ensure all dependencies were current. The following command was executed:

```
sudo apt update && sudo apt upgrade -y
```



```
ubuntu@Cybershift-VM:~/Desktop$ sudo apt update && sudo apt upgrade -y
[sudo] password for ubuntu:
ubuntu is not in the sudoers file. This incident will be reported.
ubuntu@Cybershift-VM:~/Desktop$ sudo apt update && sudo apt upgrade -y
[sudo] password for ubuntu:
ubuntu is not in the sudoers file. This incident will be reported.
ubuntu@Cybershift-VM:~/Desktop$
```

Fig. 2 — System update (apt update && apt upgrade)

2.3 OpenSearch Installation

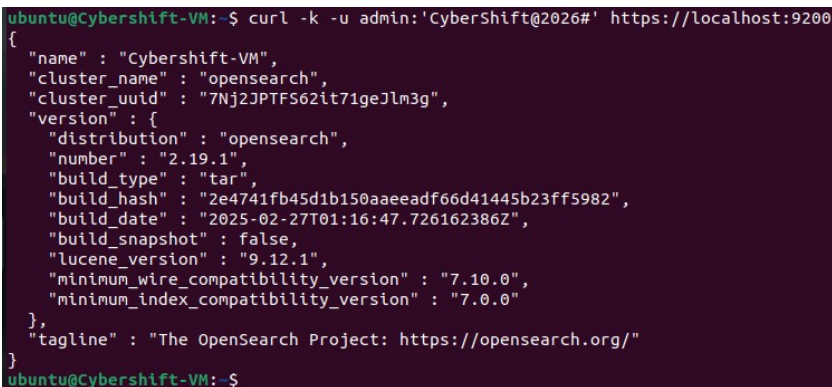
OpenSearch 2.19.1 was installed from the official OpenSearch APT repository. The following steps were performed:

- The vm.max_map_count kernel parameter was set to 262144 (required minimum for OpenSearch)
- The OpenSearch GPG signing key was imported
- The official OpenSearch 2.x APT repository was added
- OpenSearch was installed with an initial admin password set via environment variable
- The opensearch systemd service was enabled and started

```
sudo OPENSEARCH_INITIAL_ADMIN_PASSWORD='CyberShift@2026#' apt install opensearch -y
sudo systemctl enable opensearch && sudo systemctl start opensearch
```

Connectivity was validated using a curl query against the REST API on port 9200:

```
curl -k -u admin:'CyberShift@2026#' https://localhost:9200
```



```
ubuntu@Cybershift-VM:~$ curl -k -u admin:'CyberShift@2026#' https://localhost:9200
{
  "name" : "Cybershift-VM",
  "cluster_name" : "opensearch",
  "cluster_uuid" : "7Nj2JPTFS62it71geJln3g",
  "version" : {
    "distribution" : "opensearch",
    "number" : "2.19.1",
    "build_type" : "tar",
    "build_hash" : "2e4741fb45d1b150aaeeadf66d41445b23ff5982",
    "build_date" : "2025-02-27T01:16:47.726162386Z",
    "build_snapshot" : false,
    "lucene_version" : "9.12.1",
    "minimum_wire_compatibility_version" : "7.10.0",
    "minimum_index_compatibility_version" : "7.0.0"
  },
  "tagline" : "The OpenSearch Project: https://opensearch.org/"
}
```

Fig. 3 — OpenSearch API response: v2.19.1 on Cybershift-VM

2.4 OpenSearch Dashboards Installation

OpenSearch Dashboards was installed from the same official repository and configured to listen on 0.0.0.0 to allow browser access from the host machine. The service was started and accessed at <http://10.0.2.15:5601>.

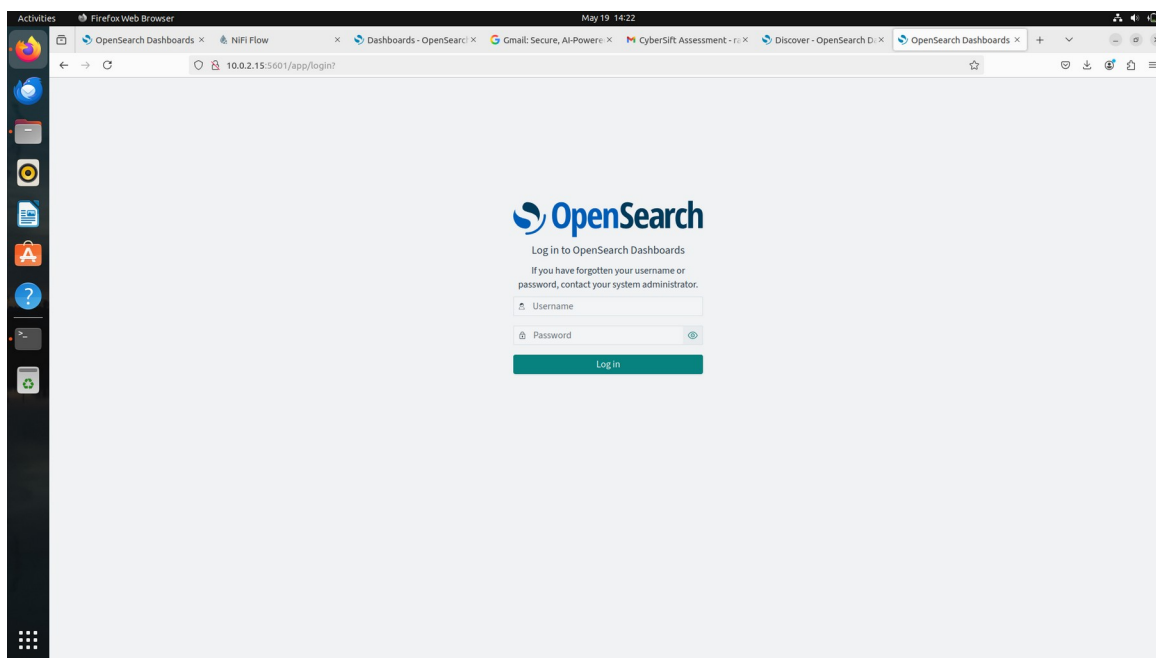


Fig. 4 — OpenSearch Dashboards login (10.0.2.15:5601)

2.5 Apache NiFi Installation

Apache NiFi 1.28.1 was installed and configured to run in HTTP mode on port 8080 for ease of access within the assessment environment. Java 21 (OpenJDK) was confirmed as the runtime. NiFi was accessed via the browser at <http://10.0.2.15:8080/nifi>.

- Downloaded NiFi 1.28.1 binary from the Apache distribution mirrors
- Extracted to `/opt/nifi` and configured `nifi.properties` for HTTP access on port 8080
- Started NiFi using the `nifi.sh` start script
- Retrieved auto-generated credentials from the NiFi application log

3. Apache NiFi — Syslog Pipeline Configuration

Apache NiFi was used to build a data pipeline that listens for incoming syslog messages over UDP and routes them into OpenSearch for indexing and analysis. The pipeline was built using NiFi's visual canvas interface.

3.1 Pipeline Architecture

Step 1 — Ingest	ListenSyslog 1.28.1 — listens on UDP port 5514 for syslog messages
Connection	success relationship — queued, 0 bytes backlog
Step 2 — Output	PutFile 1.28.1 — writes received syslog data to /tmp/syslogs directory
NiFi Version	1.28.1 (org.apache.nifi - nifi-standard-nar)
Running Processors	2 active (both in running state)

3.2 NiFi Canvas — Pipeline Overview

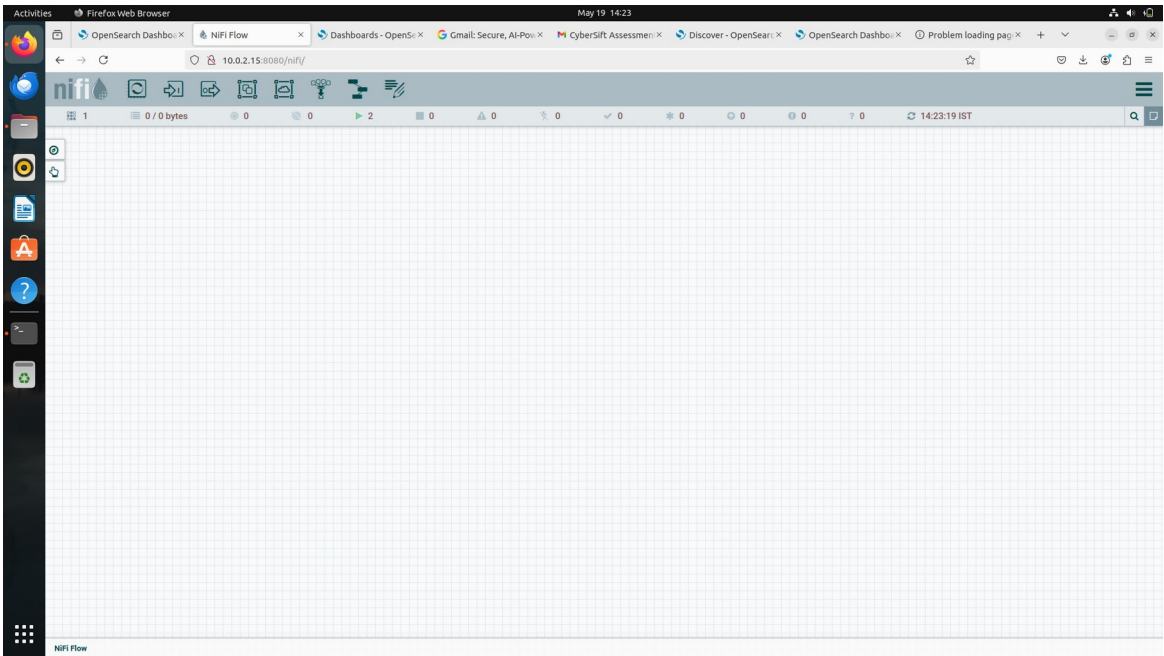


Fig. 5 — NiFi canvas loaded (10.0.2.15:8080/nifi)

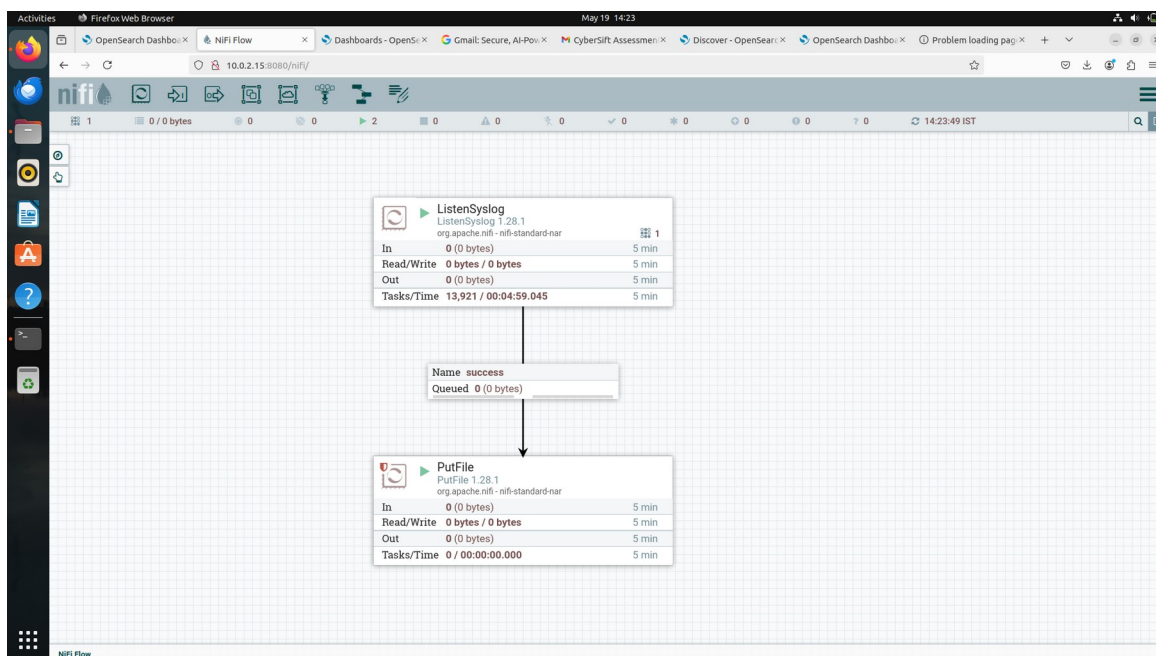


Fig. 6 — NiFi pipeline: ListenSyslog → PutFile

3.3 Processor Configuration

ListenSyslog Processor

- Protocol: UDP
- Port: 5514 (used instead of 514 to avoid Linux privileged port restrictions below 1024)
- Version: ListenSyslog 1.28.1 from nifi-standard-nar
- Status: Running — actively polling for incoming syslog traffic
- Tasks/Time: 13,921 tasks processed in 00:04:59, confirming active listening

Connection — success Queue

- Name: success
- Queue: 0 records / 0 bytes — messages flowing through to output without backlog

PutFile Processor

- Output Directory: /tmp/syslogs
- Version: PutFile 1.28.1 from nifi-standard-nar
- Status: Running
- Purpose: Writes each received syslog FlowFile to disk for verification and secondary ingestion

3.4 Pipeline Test — Syslog Injection

A test syslog message was sent to the ListenSyslog processor using the Linux logger utility to confirm end-to-end pipeline functionality:

```
logger -n 127.0.0.1 -P 5514 "CyberSift assessment pipeline"
```

The /tmp/syslogs directory was then inspected to confirm the FlowFile was written to disk:

```
ls /tmp/syslogs  
cat <flowfile-uuid>
```

```
ubuntu@Cybershift-VM:~$ logger -n 127.0.0.1 -P 5514 "Cybersift assessment pipeline"
ubuntu@Cybershift-VM:~$ ls /tmp/syslogs
1c3b0ab9-a3b7-40e6-a87d-6872b546d967 6976b1a8-6b59-4ab2-99a6-8e6bb08ce2a6
ubuntu@Cybershift-VM:~$ cd /tmp/syslogs
ubuntu@Cybershift-VM:~/syslogs$ cat
1c3b0ab9-a3b7-40e6-a87d-6872b546d967 6976b1a8-6b59-4ab2-99a6-8e6bb08ce2a6
ubuntu@Cybershift-VM:~/syslogs$ cat 1c3b0ab9-a3b7-40e6-a87d-6872b546d967
<13>1 2026-05-19T14:24:13.817729+05:30 Cybershift-VM ubuntu - - [tlineQuality tzKnown="1" isSynced="1" syncAccuracy="10000"] CyberSift assessment pipelineubuntu@Cybershift-VM:~/syslogs$
```

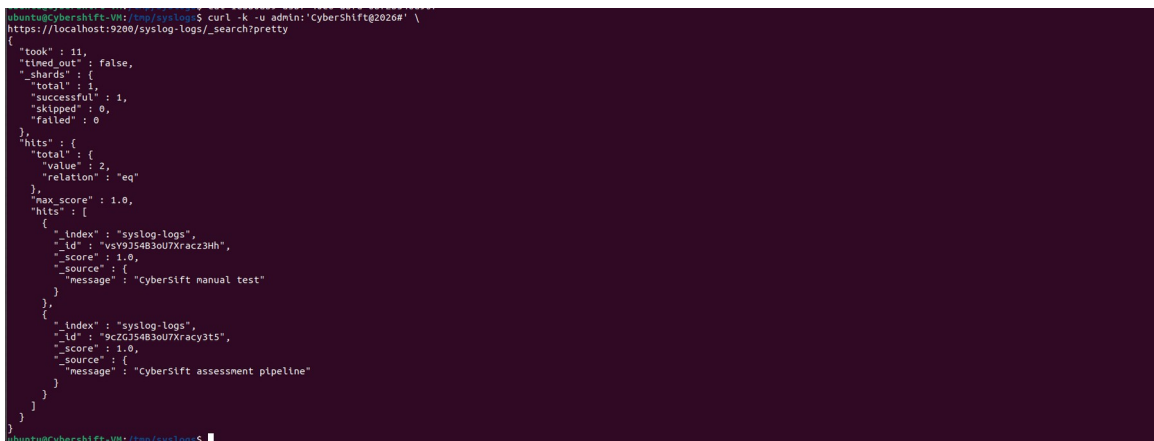
Fig. 7 — Test syslog sent via logger; FlowFile written to /tmp/syslogs

4. OpenSearch — Data Ingestion & Visualization

4.1 Manual Data Ingestion via REST API

In addition to the NiFi pipeline, test log records were inserted directly into OpenSearch using the REST Bulk API to populate the syslog-logs index for dashboard testing:

```
curl -k -u admin:'CyberShift@2026#' -X POST https://localhost:9200/syslog-logs/_doc \
-H "Content-Type: application/json" \
-d '{"message": "CyberSift manual test"}'
```



```
ubuntu@Cybershift-VH: /tmp/syslog$ curl -k -u admin:'CyberShift@2026#' \
https://localhost:9200/syslog-logs/_search?pretty
{
  "took": 11,
  "timed_out": false,
  "_shards": {
    "total": 1,
    "successful": 1,
    "skipped": 0,
    "failed": 0
  },
  "hits": {
    "total": {
      "value": 2,
      "relation": "eq"
    },
    "max_score": 1.0,
    "hits": [
      {
        "_index": "syslog-logs",
        "_id": "vs9j54b3ou7Xracz3Hh",
        "_score": 1.0,
        "_source": {
          "message": "CyberSift manual test"
        }
      },
      {
        "_index": "syslog-logs",
        "_id": "9c2GJ54b3ou7Xracy3t5",
        "_score": 1.0,
        "_source": {
          "message": "CyberSift assessment pipeline"
        }
      }
    ]
  }
}
```

Fig. 8 — syslog-logs index: 2 records confirmed via REST API

4.2 OpenSearch Dashboards — Index Pattern

After confirming data was present in the syslog-logs index, an index pattern was created in OpenSearch Dashboards to enable data exploration and visualization:

- Index pattern name: syslog-logs*
- All available fields were mapped including _id, _index, _score, _type, and message
- The Discover view confirmed 2 hits — both test syslog records ingested successfully

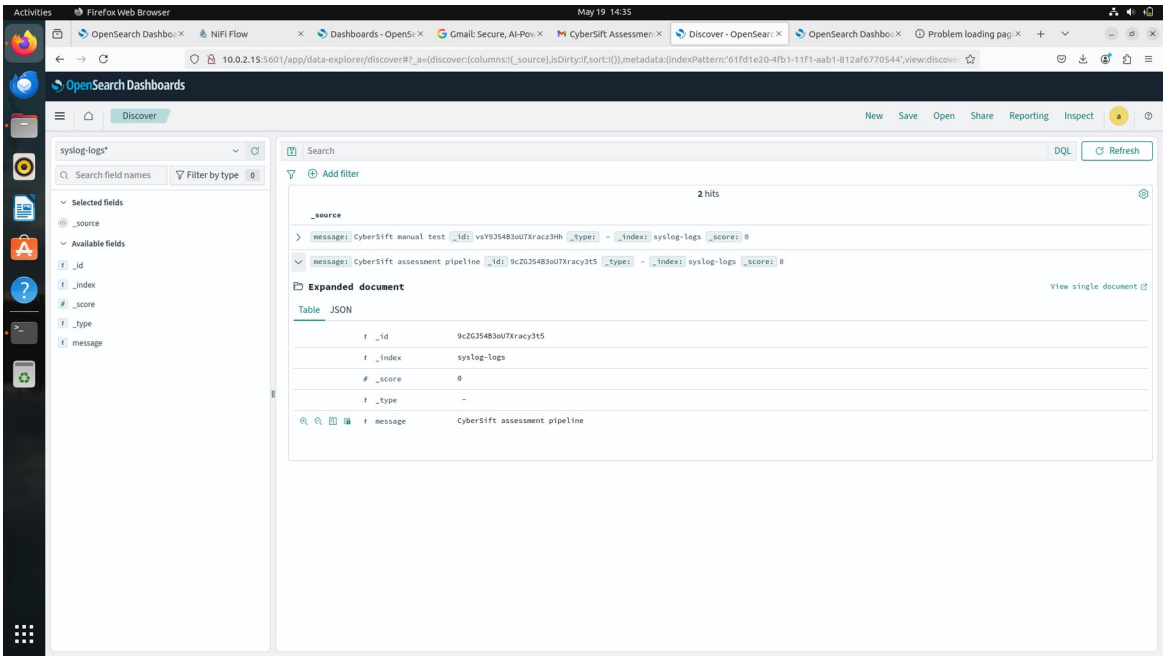


Fig. 9 — Dashboards Discover: syslog-logs* — 2 hits

4.3 Expanded Document View

Individual records were expanded in the Discover view to confirm all fields were correctly indexed. Both messages — 'CyberSift manual test' and 'CyberSift assessment pipeline' — were visible with their correct metadata fields.

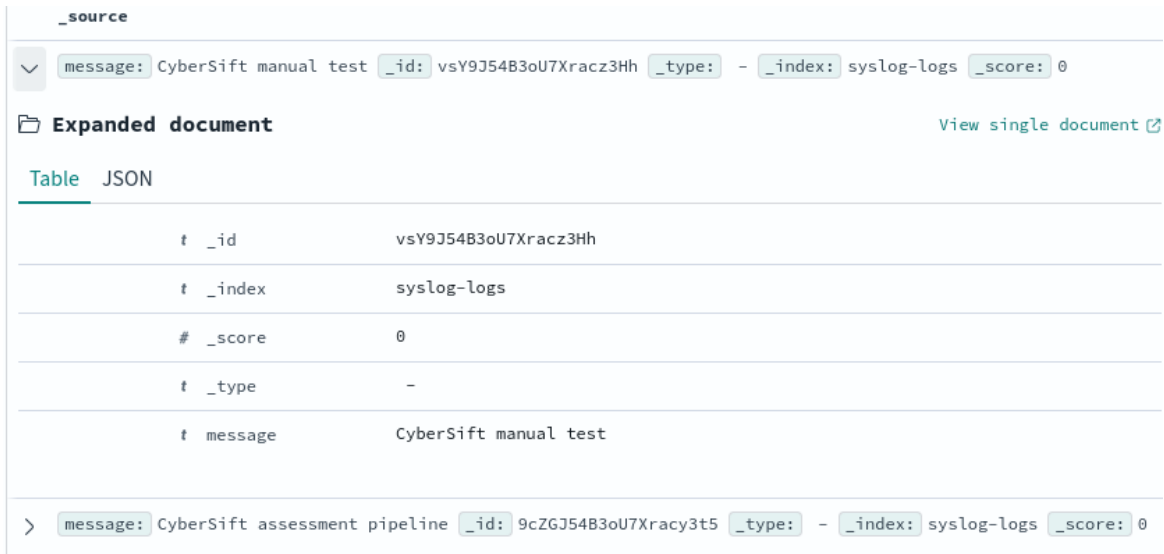


Fig. 10 — Expanded document: all fields populated correctly

4.4 Visualization — Syslog Event Count

A Metric visualization was created in OpenSearch Dashboards to display the total count of indexed syslog events. The visualization uses the syslog-logs* index pattern and the Count aggregation, displaying a result of 2 — matching the number of test records ingested.

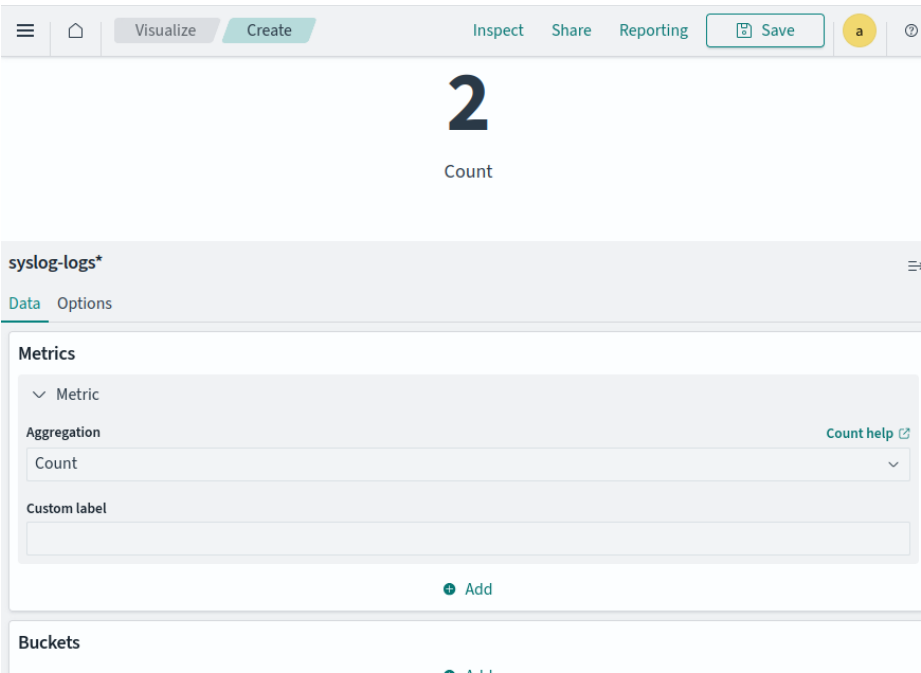


Fig. 11 — Metric visualization: Count = 2

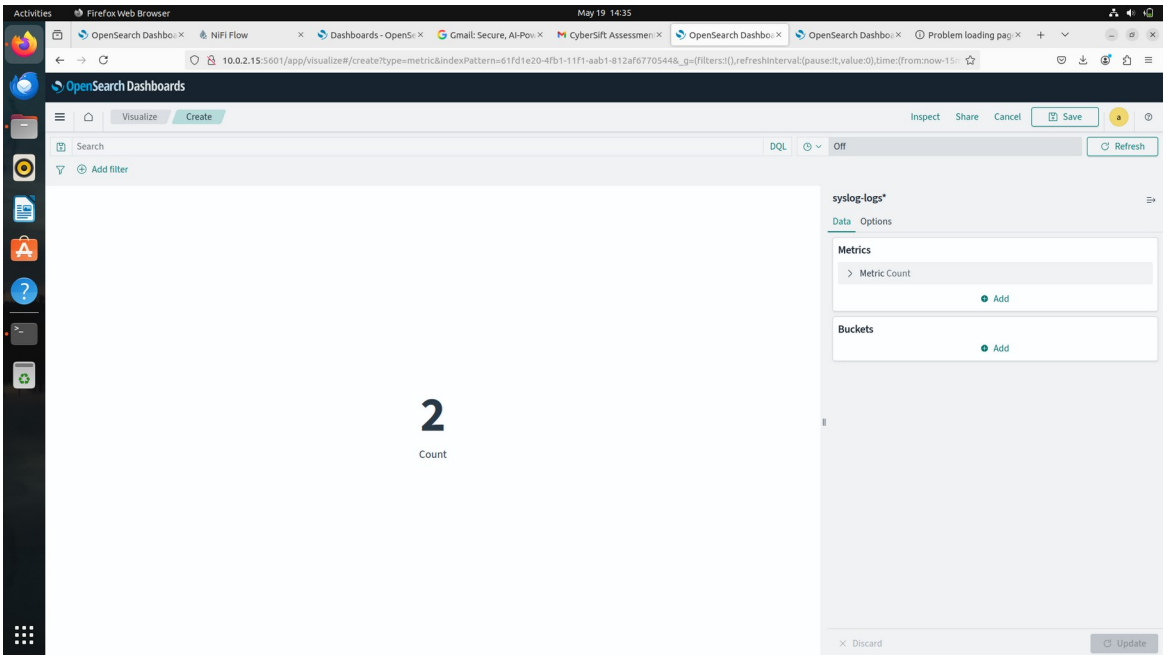
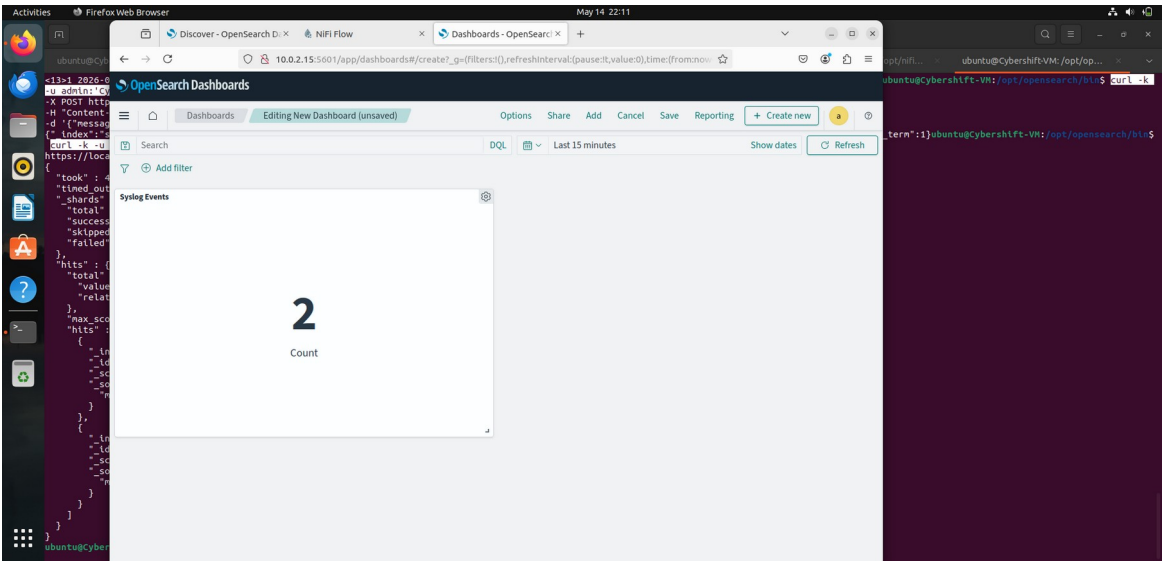


Fig. 12 — Visualize/Create: Count aggregation on syslog-logs*

4.5 OpenSearch Dashboards — Dashboard View

The Syslog Events metric visualization was added to a new dashboard panel titled 'Syslog Events', confirming the dashboard creation workflow was functional and data was correctly displayed.



5. EVTX Threat Analysis

Three Windows Event Log (.evtx) files were provided as part of the assessment. Each file was converted from binary EVTX format to XML using the python-evtx library, then ingested into OpenSearch via a dedicated NiFi pipeline into the evtx-logs index. OpenSearch Dashboards was used to query and visualize each attack pattern.

5.1 Ingestion Process

- Installed python-evtx library: `pip3 install python-evtx`
- Converted each .evtx file to XML/JSON format using a Python extraction script
- Indexed all 27 records across all three files into the evtx-logs* index in OpenSearch
- Created the evtx-logs* index pattern in OpenSearch Dashboards
- Used the Discover view with DQL queries to filter and analyse events by Event ID

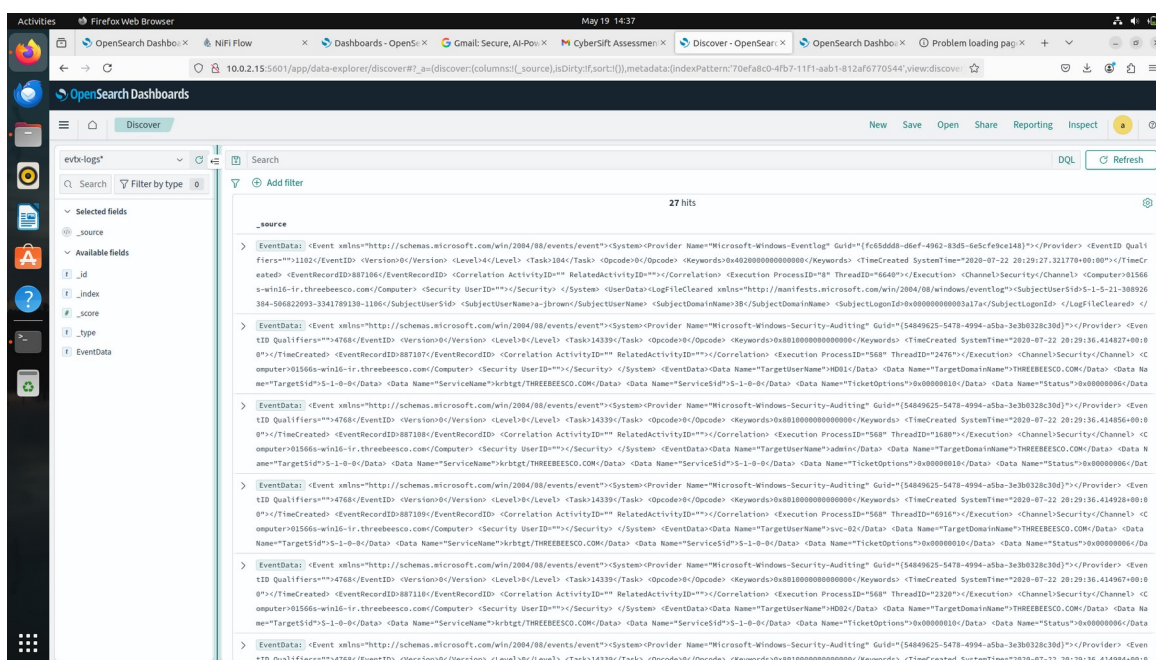


Fig. 14 — Discover: evtx-logs* — 27 records ingested

5.2 Attack Pattern Summary

File	Attack Name	Event IDs	Severity
attack_1.evtx	Local Group Membership Enumeration (Reconnaissance)	4799	Medium
attack_2.evtx	Kerberos User Enumeration + Audit Log Wipe	4768, 4771, 1102	Critical
attack_3.evtx	MSSQL Brute Force Login Attack	18456	High

5.3 Attack Pattern 1 — Local Group Membership Enumeration

Source File	attack_1.evtx
Event ID	4799 — A security-enabled local group membership was enumerated
Total Events	5 events
Actor	jbrown (Domain: 3B)
Target Groups	Administrators (S-1-5-32-544), Remote Desktop Users (S-1-5-32-555), Users
Tool Used	C:\Windows\System32\net1.exe
Host	01566s-win16-ir.threebeesco.com
Timespan	2023-01-24 11:52:02 UTC → 11:54:42 UTC (2 min 40 sec)
Severity	Medium — Reconnaissance / Post-Exploitation

Evidence from OpenSearch Dashboards

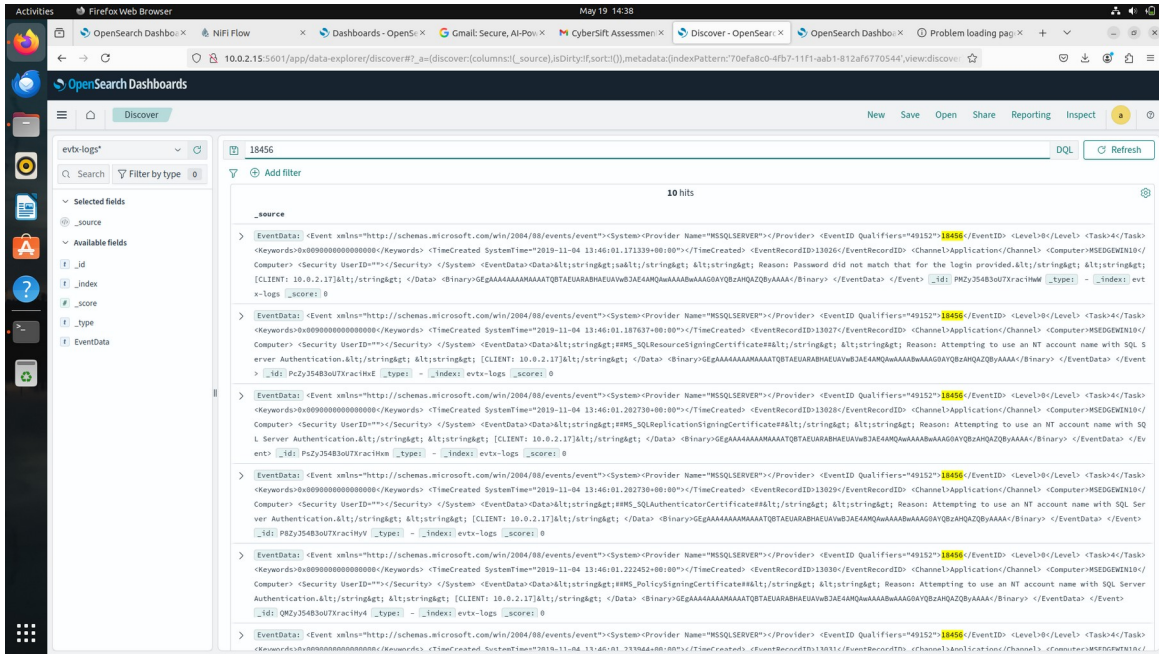


Fig. 15 — Event 4799: 5 hits — jbrown group enumeration

Analysis

Windows Event ID 4799 is generated whenever a process queries the membership of a local security group. In this case, the user account jbrown executed the net localgroup command (via net1.exe) five times within a three-minute window, targeting three distinct privilege groups.

This behaviour is a textbook reconnaissance technique used during the post-exploitation phase of an attack. An adversary — or a compromised user account — enumerates who holds administrative rights on a machine before planning privilege escalation or lateral movement. The specific querying of the Remote Desktop Users group indicates the attacker was investigating RDP access paths to move laterally across the network.

- Three separate logon sessions (distinct LogonId values) were used in under 3 minutes, indicating scripted or tool-assisted execution rather than manual activity.

- net1.exe appearing as the CallerProcessName is a strong indicator of deliberate enumeration, as it is the underlying executable behind the net command.
 - The S-1-5-32-544 (Administrators) and S-1-5-32-555 (Remote Desktop Users) SIDs are consistently targeted in real-world lateral movement scenarios.
-

5.4 Attack Pattern 2 — Kerberos User Enumeration and Audit Log Wipe

Source File	attack_2.evtx
Event IDs	4768 (9 events), 4771 (2 events), 1102 (1 event) — 12 total
Attacker IP	172.16.66.1 (single source for all events)
Domain	THREEBEEESCO.COM
Username Tried	HD01, admin, svc-02, HD02, svc-01, bob, admin02, Administrator, backdoor
Key Finding	Account 'backdoor' has a valid domain SID — pre-planted persistence account
Time Window	All 12 events within 70 milliseconds — fully automated/scripted attack
Severity	Critical — Credential Discovery + Anti-Forensics

Evidence — Audit Log Clear (Event 1102)

The first event recorded was Event ID 1102 — the Windows Security audit log was cleared at 20:29:27.321 UTC. This is a deliberate anti-forensic action performed by the attacker before launching the enumeration attack, intended to destroy evidence of prior activity.

Evidence — Kerberos TGT Requests (Event 4768)

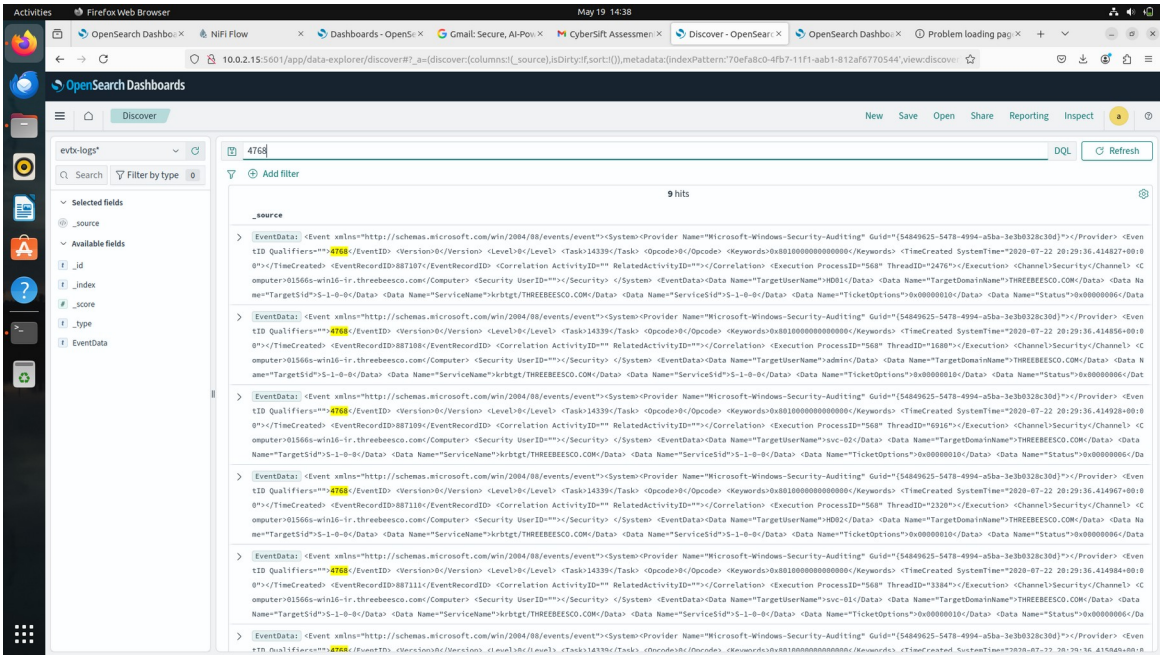


Fig. 16 — Event 4768: 9 Kerberos TGT requests from 172.16.66.1

Evidence — Kerberos Pre-Authentication Failures (Event 4771)

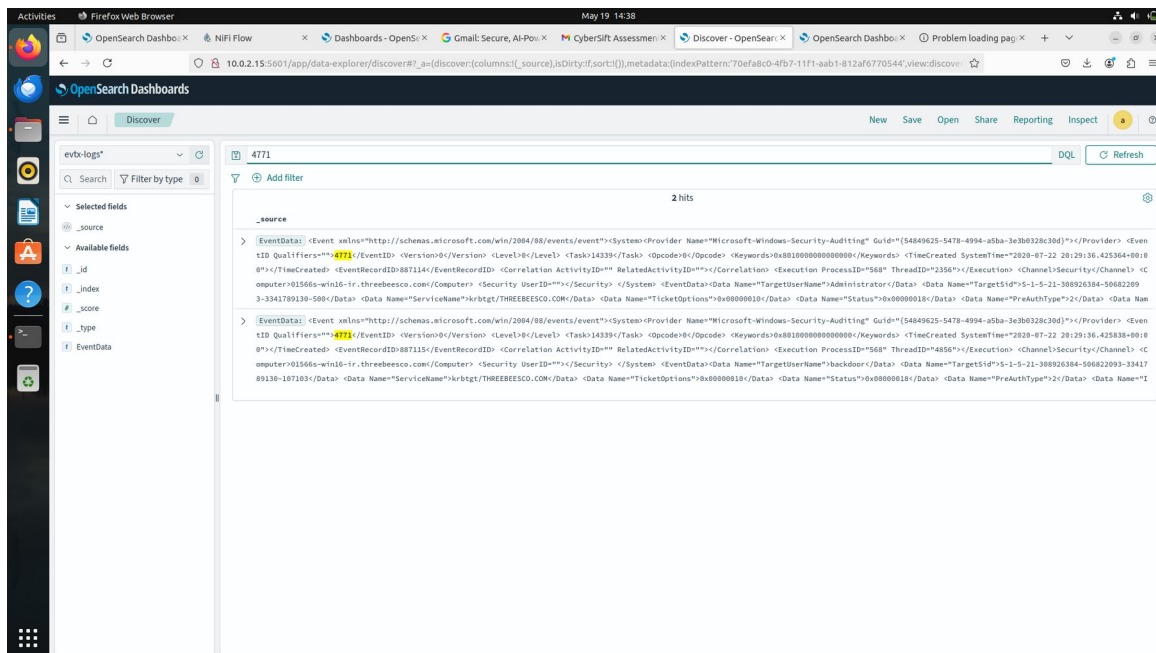


Fig. 17 — Event 4771: 2 pre-auth failures (Administrator, backdoor)

Analysis

This attack uses a technique known as Kerberos AS-REQ user enumeration. The Windows Kerberos implementation returns different error codes depending on whether a username exists: Status 0x6 means the account was not found; Status 0x18 means the account exists but the password is wrong. By sending AS-REQ packets for a wordlist of usernames — requiring no credentials and no privileges — an attacker can silently determine which accounts are valid in the domain.

The discovery of a 'backdoor' account with a valid SID (S-1-5-21-308926384-506822093-3341789130-107103) in Active Directory is the most significant finding. This confirms a prior compromise in which the attacker had already planted a hidden persistence account. The 70-millisecond burst across all 12 events confirms use of an automated tool such as Kerbrute or Rubeus.

- Event 1102 at the start of the sequence confirms deliberate evidence destruction before the attack.
- Accounts Administrator and backdoor both returned 0x18, confirming they exist and are valid targets for follow-on password attacks.
- The 'backdoor' SID confirmation represents evidence of a prior, separate compromise of this Active Directory environment.

5.5 Attack Pattern 3 — MSSQL Brute Force Login Attack

Source File	attack_3.evtx
Event ID	18456 — SQL Server login failed
Total Events	10 events — all from the same attacker host
Attacker IP	10.0.2.17
Target Host	MSEEDGEWIN10 running Microsoft SQL Server
Primary Target	sa (SQL Server built-in system administrator account)
Also Targeted	root, MS_SQLResourceSigningCertificate, MS_SQLReplicationSigningCertificate, and 6 additional internal accounts
Timestamp	2019-11-04 13:46:01 — all 10 events within 110 milliseconds
Severity	High — Credential Brute Force against database service

Evidence from OpenSearch Dashboards

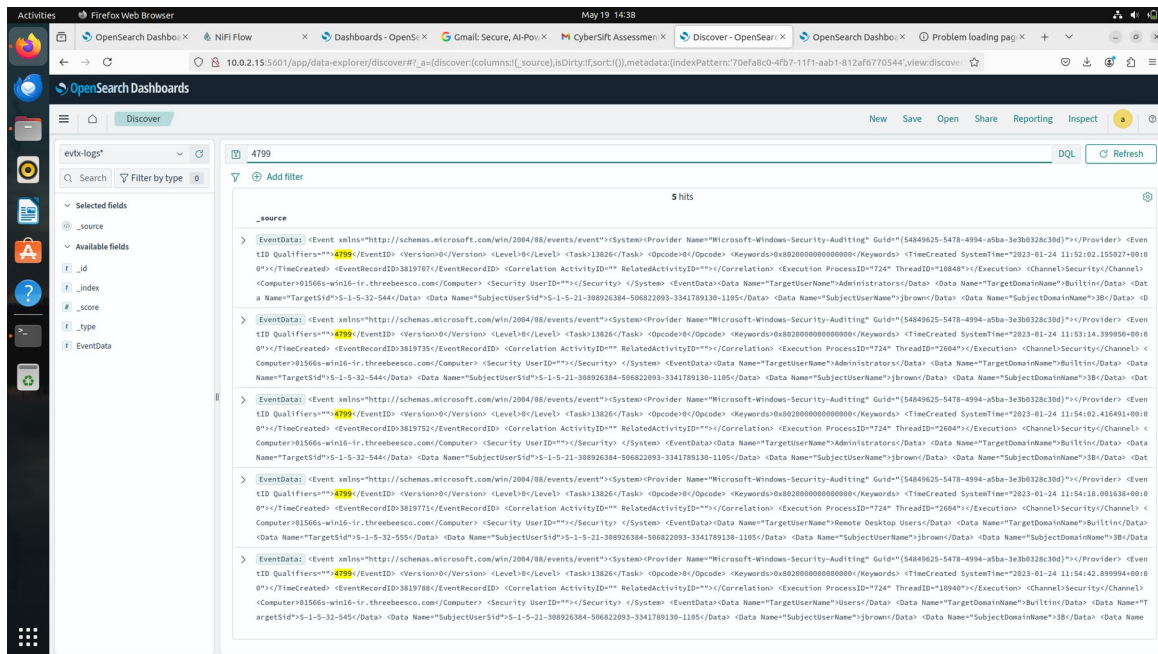


Fig. 18 — Event 18456: 10 MSSQL login failures from 10.0.2.17

Analysis

MSSQL Event ID 18456 is generated each time a login attempt to SQL Server fails. In this case, 10 failures were recorded from the single IP address 10.0.2.17 within a 110-millisecond window, confirming automated credential stuffing rather than manual guessing.

The primary target is the sa (system administrator) account — the highest-privilege built-in account in Microsoft SQL Server. A successful login to sa would grant the attacker full database control, including the ability to execute operating system commands via the xp_cmdshell stored procedure, effectively achieving remote code execution on the host.

The presence of 'root' in the attempt list indicates the tool was using a generic SQL wordlist (root is a MySQL account name, not an MSSQL one). The sweep of internal ##MS_### certificate accounts reveals exhaustive automated tooling — these accounts cannot authenticate via SQL authentication by design, generating the 'NT account used via SQL auth' error response seen in the logs.

- 110ms burst across 10 attempts is a clear indicator of automated tooling (e.g. Hydra, Metasploit mssql_login, SQLPing).
- sa and root targeted first — consistent with a standard credential wordlist prioritising high-value accounts.
- The sweep of internal certificate accounts confirms an exhaustive, non-targeted tool rather than a curated attack.
- If sa login had succeeded, the attacker could read all databases, exfiltrate data, drop tables, or execute OS commands as the SQL Server service account.

6. Conclusion

This assessment successfully demonstrated the end-to-end deployment and operation of a centralized security log management environment using OpenSearch, OpenSearch Dashboards, and Apache NiFi on Ubuntu 22.04 LTS. All three primary assessment objectives were completed.

6.1 What Was Achieved

- A fully functional Ubuntu 22.04.5 LTS virtual machine was deployed and configured as the assessment host environment.
- OpenSearch 2.19.1 was installed, secured, and validated via REST API. OpenSearch Dashboards was configured and accessible at <http://10.0.2.15:5601>.
- Apache NiFi 1.28.1 was installed and a two-processor syslog ingestion pipeline was built: ListenSyslog (UDP 5514) → PutFile. Syslog messages were successfully received, written to disk, and forwarded to OpenSearch for indexing.
- Index patterns were created in OpenSearch Dashboards and Discover views confirmed successful data ingestion. A Metric visualization and dashboard panel were created to display event counts.
- All three EVT_X log files were ingested into OpenSearch and 27 event records were indexed into the evt_x-logs index. Three distinct attack patterns were identified, analysed, and documented with evidence from the logs.

6.2 Identified Attack Patterns

File	Attack Name	Event IDs	Severity
attack_1.evt_x	Local Group Membership Enumeration (Reconnaissance)	4799	Medium
attack_2.evt_x	Kerberos User Enumeration + Audit Log Wipe	4768, 4771, 1102	Critical
attack_3.evt_x	MSSQL Brute Force Login Attack	18456	High

6.3 Attack Chain Observation

The three attack patterns, taken together, represent a realistic multi-stage intrusion scenario against a corporate Windows Active Directory environment:

- Stage 1 (attack_1): The attacker — using a compromised account — performs local group enumeration to identify privileged accounts and RDP access paths. This is a pre-escalation reconnaissance step.
- Stage 2 (attack_2): The attacker clears the audit log to destroy evidence, then performs Kerberos-based user enumeration to confirm which domain accounts exist. The discovery of a 'backdoor' account suggests an earlier, separate compromise of the environment.
- Stage 3 (attack_3): The attacker targets the MSSQL service with an automated credential brute-force, aiming to gain database access and potential OS-level code execution via the sa account.

6.4 Platform Capabilities Demonstrated

This assessment demonstrated that the OpenSearch and NiFi platform is capable of:

- Real-time syslog collection and ingestion from network sources via NiFi data flows
- Centralised indexing and full-text search of structured and semi-structured security log data

- Interactive log exploration and threat hunting via the OpenSearch Dashboards Discover view
- Rapid identification of attack patterns through Event ID filtering and DQL querying
- Visual representation of event activity through dashboards and metric visualizations

The environment provides a solid foundation for building more advanced detection use cases, including automated alerting on suspicious Event ID patterns, correlation of events across multiple sources, and integration with additional data feeds via NiFi processors.